

Deformation History Through Reconnection

Why direct full-gradient transport is preferable to $\mathbf{T}_{\text{total}}$

Technical note for the MTS2D reconnection model

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1 Two different histories

For the currently selected local reference triangle,

$$\mathbf{F} = \mathbf{F}_E \mathbf{F}_P. \quad (1)$$

The existing plastic history is

$$\mathbf{T} = \mathbf{F}_P \mathbf{H}_P, \quad \mathbf{T}_{\text{total}} := \mathbf{F}_E \mathbf{T} = \mathbf{F} \mathbf{H}_P. \quad (2)$$

At a flip, the update

$$\mathbf{H}_P^+ = (\mathbf{F}_P^+)^{-1} \mathbf{F}_P^- \mathbf{H}_P^- \quad (3)$$

preserves the plastic branch exactly, $\mathbf{T}^+ = \mathbf{T}^-$. The reconstructed full quantity is not preserved:

$$\boxed{\mathbf{T}_{\text{total}}^+ - \mathbf{T}_{\text{total}}^- = (\mathbf{F}_E^+ - \mathbf{F}_E^-) \mathbf{T}^-}. \quad (4)$$

This is appropriate if $\mathbf{T}$ is interpreted only as plastic bookkeeping. It is problematic if $\mathbf{T}_{\text{total}}$ is interpreted as the total deformation experienced by the material, because a refreshed local elastic factor can modify previously stored deformation.

Direct transport instead defines

$$\mathcal{F} = \mathbf{F} \mathbf{H}_F, \quad \mathbf{H}_F^+ = (\mathbf{F}^+)^{-1} \mathbf{F}^- \mathbf{H}_F^-. \quad (5)$$

Therefore

$$\boxed{\mathcal{F}^+ = \mathbf{F}^+ \mathbf{H}_F^+ = \mathbf{F}^- \mathbf{H}_F^- = \mathcal{F}^-}. \quad (6)$$

The mesh still selects a new local square reference after reconnecting; the original triangle is not retained as the working reference of $\mathbf{F}$. However, $\mathbf{H}_F$ accumulates the inverse changes of local reference. Consequently, $\mathcal{F} = \mathbf{F} \mathbf{H}_F$ behaves exactly as if one original material reference had been kept throughout. Elastic and plastic deformation both remain in the total history, while the constitutive energy may continue to use the refreshed local $\mathbf{F}_E$.

2 A decisive simple-shear check

Consider a periodic cell with imposed simple shear

$$\bar{\mathbf{F}}(\gamma) = \begin{pmatrix} 1 & \gamma \\ 0 & 1 \end{pmatrix}. \quad (7)$$

For a deformation $\mathbf{y}(\mathbf{X}) = \bar{\mathbf{F}}\mathbf{X} + \mathbf{w}$ with periodic $\mathbf{w}$, compatibility requires

$$\langle \nabla_X \mathbf{y} \rangle = \bar{\mathbf{F}} + \langle \nabla_X \mathbf{w} \rangle = \bar{\mathbf{F}}. \quad (8)$$

The periodic contribution averages to zero. Thus any field interpreted as the total deformation gradient from the original material reference must satisfy

$$\langle F_{12} \rangle = \gamma, \quad \langle F_{21} \rangle = 0, \quad (9)$$

regardless of internal plastic slips or edge flips.

In the 70×70 periodic test, both histories follow the imposed shear until the intensive reconnection regime near $\gamma = 0.72$. At $\gamma = 1$,

$$\langle \mathcal{F} \rangle \simeq \begin{pmatrix} 1.006 & 1.005 \\ -0.001 & 1.009 \end{pmatrix}, \quad \langle \mathbf{T}_{\text{total}} \rangle \simeq \begin{pmatrix} 0.957 & 0.335 \\ 0.493 & 0.990 \end{pmatrix}. \quad (10)$$

The direct history reproduces the imposed map to about one percent. $\mathbf{T}_{\text{total}}$, by contrast, loses two thirds of the imposed 12 shear and develops a large 21 component that is absent from the boundary deformation. This is not a different mechanical response: $\mathcal{F}$ was passive, so both fields were measured on the same simulation trajectory. The discrepancy is entirely a difference in history interpretation.

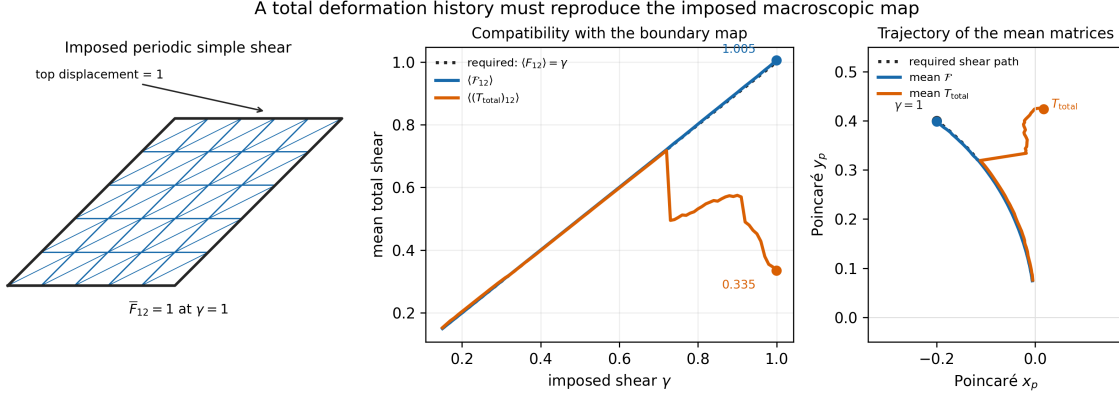


Figure 1: Macroscopic compatibility provides an objective comparison. The center panel shows that $\langle \mathcal{F}_{12} \rangle$ follows the imposed shear, while $\langle (\mathbf{T}_{\text{total}})_{12} \rangle$ turns away after intensive reconnecting. The right panel shows the same result in the Poincaré disk: the trajectory of the mean $\mathcal{F}$ remains on the required pure-shear path, whereas the mean $\mathbf{T}_{\text{total}}$ leaves it.

3 Advantages and limitations

	Plastic reconstruction $\mathbf{T}_{\text{total}}$	Direct history $\mathcal{F}$
Exactly preserved	Plastic branch $\mathbf{T}$.	Total gradient $\mathcal{F}$.
Effective reference	Refreshed $\mathbf{F}_E$ combined with the stored plastic branch.	Original material reference carried implicitly by $\mathbf{H}_F$.
Elastic deformation	Only the current local elastic factor is included.	All accumulated elastic and plastic deformation is retained.
After many flips	Components can be reinterpreted through Equation (4).	Unchanged by the flip itself; changes only with subsequent physical deformation.
Simple-shear result	Violates the imposed macroscopic deformation when read as a total gradient.	Satisfies the macroscopic compatibility condition.
Best use	$\mathbf{T}$ remains valuable for identifying plastic slips, but $\mathbf{T}_{\text{total}}$ should not be used as total material history.	Primary total-history field for output, trajectories, and Poincaré analysis.

Both approaches still require a rule assigning the histories of two parent triangles to the two children created by an edge flip. That remeshing choice is shared and is not a reason to prefer either formula. The relevant distinction is what is preserved once the parent-child pairing has been chosen: only $\mathcal{F}$ preserves the complete deformation history.

4 Recommendation

Use $\mathcal{F} = \mathbf{F}\mathbf{H}_F$ as the total deformation-history field, while retaining the existing local reduction and $\mathbf{F}_E$ in the constitutive energy. Keep $\mathbf{T}$ as the plastic diagnostic. Then $\mathbf{F}_E$ means current recoverable deformation, $\mathbf{T}$ means accumulated plastic lattice branch, and $\mathcal{F}$ means total deformation from the effective original material reference. The closest square reference remains a good local gauge, but the reported total history no longer depends on which square candidate is selected at a flip.